David Friedenberg

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Inspire-HEP

Research Interests

Nuclear many-body theory, Neutron stars, Core-collapse supernovae, Equation of state of neutron-rich matter, Chiral effective field theory, Nucleon self-energies and spectral properties, Machine learning methods for nuclear astrophysics, Computational physics.

Education

Texas A&M University — Aug. 2021 – Expected Aug. 2027
Ph.D., Physics — College Station, TX
Advisor: Jeremy W. Holt

Texas A&M University — Aug. 2021 – Dec. 2025
M.S., Physics — College Station, TX
Thesis Title: *Understanding Neutron Stars and Supernovae with Chiral Effective Field Theory*

Yeshiva University — Aug. 2016 – June 2021
M.A., Physics (2020 – 2021) — New York, NY
B.A., Physics and Mathematics (2016 – 2020)

Research Experience

Graduate Researcher — Aug. 2022 – Present
Texas A&M University, Cyclotron Institute — College Station, TX

- Developed ab initio calculations of hot, dense nuclear matter using chiral effective field theory with finite-temperature many-body perturbation theory. This includes calculating thermodynamically consistent nuclear equations of state, nucleon self-energies, and derived quantities such as spectral functions, effective masses, and nuclear mean fields.
- Collaborated across several institutions to investigate nuclear-matter properties and their astrophysical implications for neutron star structure (MUSES Collaboration), neutron star merger remnant cooling (Mark Alford, Wash U. St. Louis), nuclear liquid-gas phase transition (Jamie Karthein, Texas A&M), and neutrino luminosity in core-collapse supernovae (William Newton, East Texas A&M).
- Developed numerical software in C++ for calculating finite temperature many-body observables for a range of nuclear interaction potentials and running parallel computations on HPC systems.
- Developed normalizing flow machine learning models in PyTorch to accelerate nuclear many-body calculations of equations of state and nucleon self-energies.

Research Experience (continued)

Developer, NSF MUSES Collaboration Aug. 2022 – Aug. 2026

- Developed a complete C++/Python module for the MUSES Neutron Stars working group to calculate the nuclear equation of state at zero and finite temperatures using chiral effective field theory. Containerized the module with Docker for deployment in HPC environments using Kubernetes.
- Developed and tested key components of the MUSES cyberinfrastructure, including an interactive graphical interface for configuring and executing scientific workflows. The interface enables users to connect modules, specify input parameters, and launch calculations of nuclear equations of state and astrophysical observables.

Publications

Peer-Reviewed Publications

(1) "Building neutron stars with the MUSES calculation engine",
M. R. Pelicer, N. Cruz-Camacho, C. Conde-Ocazionez, **D. Friedenberg**, et al.
Physical Review D 111, 103037 (2025). doi: *10.1103/PhysRevD.111.103037*

(2) "Theoretical and experimental constraints for the equation of state of dense and hot matter",
R. Kumar, V. Dexheimer, J. Jahan, ..., **D. Friedenberg**, et al.
Living Reviews in Relativity 27, 3 (2024). doi: *10.1007/s41114-024-00049-6*.

(3) "Euler–Bernoulli theory accurately predicts atomic force microscope cantilever shape during non-equilibrium snap-to-contact motion",
S. J. Eppell, **D. Friedenberg**, O. Payton, L. Picco, F. R. Zypman
Nanotechnology 31, 185702 (2020). doi: *10.1088/1361-6528/ab6dff*.

Submitted for Peer Review

(1) "Studying the QCD Matter produced in Heavy-Ion Collisions using the MUSES Calculation Engine",
J. Jahan, K. P. Pala, Y. Yang, ..., **D. Friedenberg**, et al.
arXiv:2606.26326 (2026). doi: *10.48550/arXiv.2606.26326*.

In Progress

(1) "Nucleon self-energies in hot, dense nuclear matter from chiral effective field theory",
D. Friedenberg, J. W. Holt

(2) "Critical properties of nuclear liquid-gas phase transition from chiral effective field theory",
R. Chhabra, **D. Friedenberg**, J. W. Holt, J. M. Karthein

(3) "Microscopic constraints on neutrino emissivities in neutron-rich matter",
R. L. Preston, **D. Friedenberg**, W. G. Newton, J. W. Holt

(4) "Urca rates from ab initio nucleon widths in the nucleon width approximation",
A. Haber, **D. Friedenberg**, M. G. Alford, J. W. Holt, C. Drischler

Publications (continued)

Open Source Software

(1) "Chiral EFT Equation of State module (Version 1.0.1)",
D. Friedenberg, J. W. Holt
MUSES Collaboration, Zenodo (2024). doi: *10.5281/zenodo.14611275*.

Research Presentations

Oral Presentations

(1) "Microscopic modeling of hot, dense nuclear matter in core-collapse supernovae",
Institute for Nuclear Theory Workshop INT-26-96W, *Finite-Temperature Effects in Multi-Messenger Astrophysics*, Seattle, WA (June 2026)

(2) "Microscopic Properties of Hot, Dense Nuclear Matter from Chiral EFT for Core-Collapse Supernovae",
APS Global Summit, Denver, CO (March 2026)

(3) "Nuclear Effective Masses in Hot, Dense Nuclear Matter from Chiral Effective Field Theory",
Texas, Arkansas, Oklahoma Section APS Meeting, Ft. Worth, TX (October 2025)

(4) "Chiral EFT Equation of State Module",
MUSES Collaboration Meeting, Urbana, IL (May 2023)

Poster Presentations

(1) "Nucleon Self-Energies in Hot, Dense Nuclear Matter from Chiral Effective Field Theory",
CeNAM Frontiers in Nuclear Astrophysics Meeting, College Station, TX (May 2026)

(2) "Neutron Star Equation of State from Chiral EFT within the MUSES collaboration",
NNSA CENTAUR SAC Meeting, Livermore, CA (July 2024)

Awards & Fellowships

TAOAPS Graduate Oral Presentation Award
Texas, Arkansas, Oklahoma Chapter – American Physical Society 2025

Philip and Doris Moses Fellowship
Texas A&M University 2021

Bertha Kressel Research Scholarship
Yeshiva University 2018

Teaching & Research Mentoring

Undergraduate Research Mentor Summer 2026 – Present
Texas A&M University College Station, TX

- Mentored undergraduate REU student Riddhi Chhabra, studying critical behavior of the nuclear liquid-gas phase transition using a microscopic nuclear equation of state. This project was presented at the APS Division of Nuclear Physics meeting.
- Mentored undergraduate research student Caleb Brown, accelerating ab initio calculations of the nucleon self-energy in hot, dense nuclear matter using machine learning techniques.

Graduate Teaching Assistant Fall 2021 – Spring 2022
Dept. of Physics & Astronomy, Texas A&M University College Station, TX

Recitation leader for calculus-based Mechanics (PHYS 206) and Electricity & Magnetism (PHYS 207) courses.

Adjunct Instructor, Math Department Fall 2020 – Spring 2021
Mathematics Dept., Yeshiva University New York, NY

Principal lecturer for Calculus I, Linear Algebra, and Mathematical Statistics courses, developing all course notes and leading classroom instruction.

Service & Outreach

Organizing Committee, CeNAM Frontiers in Nuclear Astrophysics 2026 May 2026
Texas A&M University College Station, TX

- Co-organizer for the CeNAM Frontiers international nuclear astrophysics meeting with 130+ participants.

Discover, Explore and Enjoy Physics (DEEP) Mentorship Program Fall 2022 – Spring 2024
Texas A&M University College Station, TX

- Worked side-by-side with undergraduate students designing and building physics demonstrations such as a Tesla coil, a magnetohydrodynamic drive, and a cloud chamber for the Texas A&M Physics & Engineering Festival and other outreach programs.